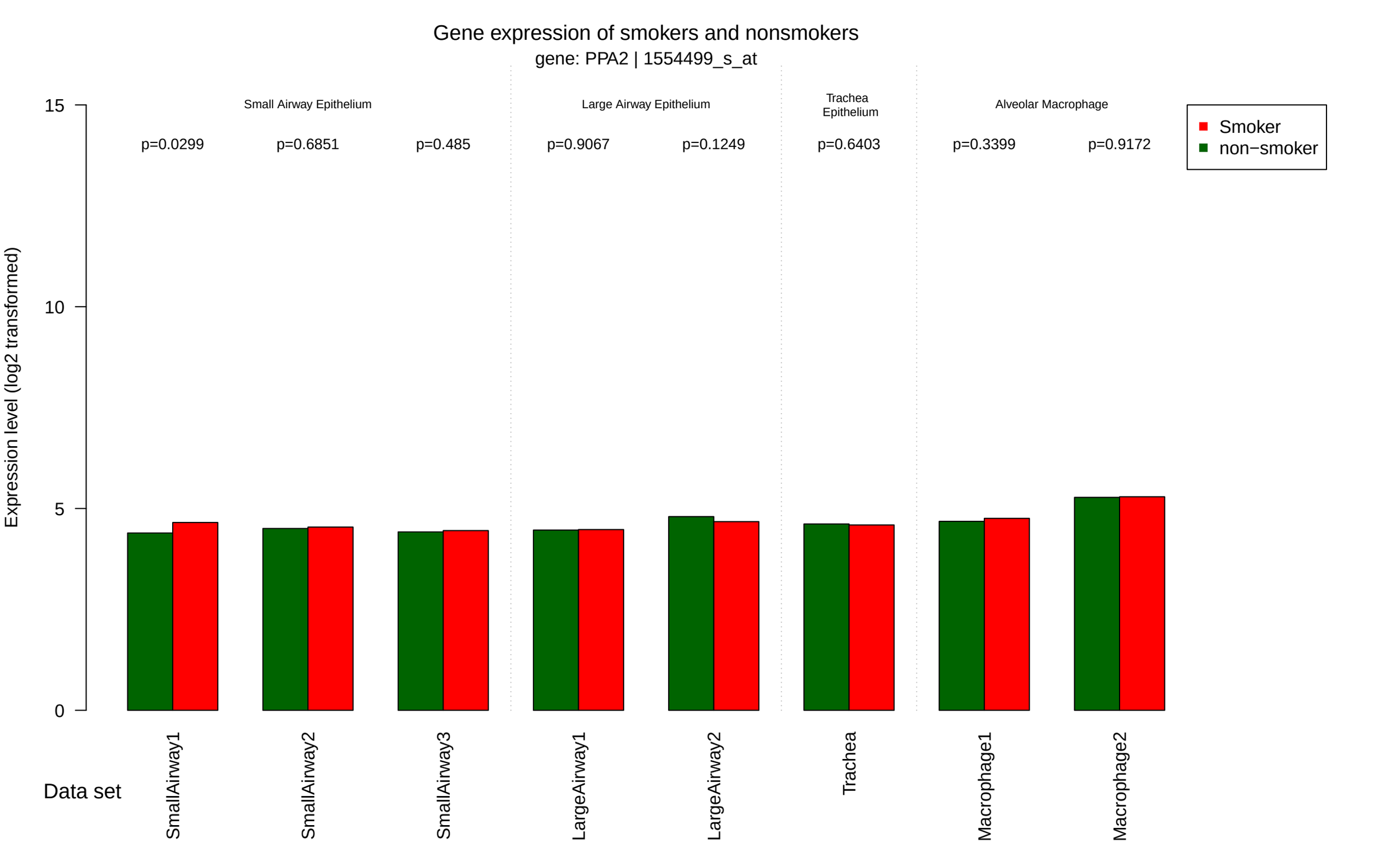
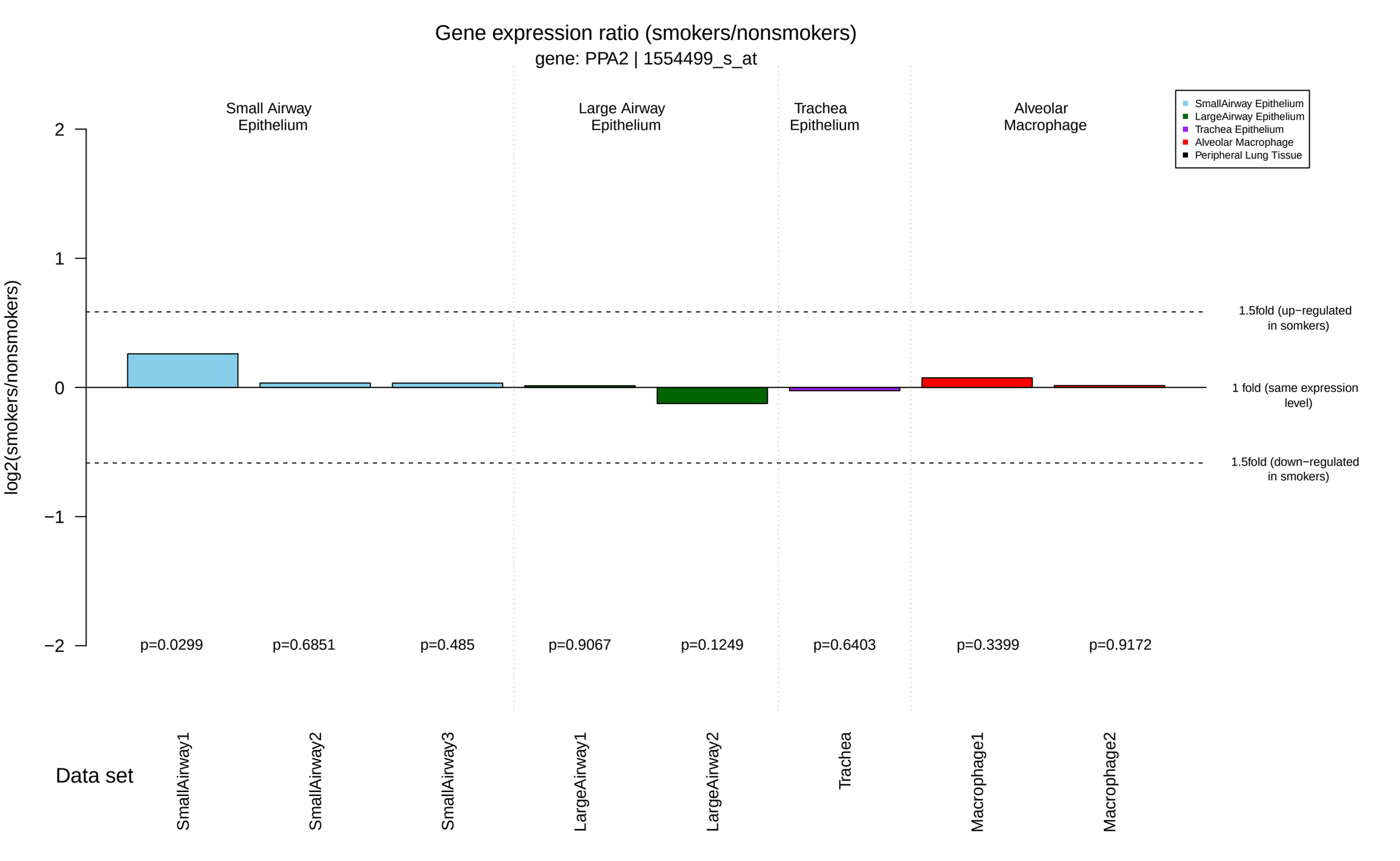
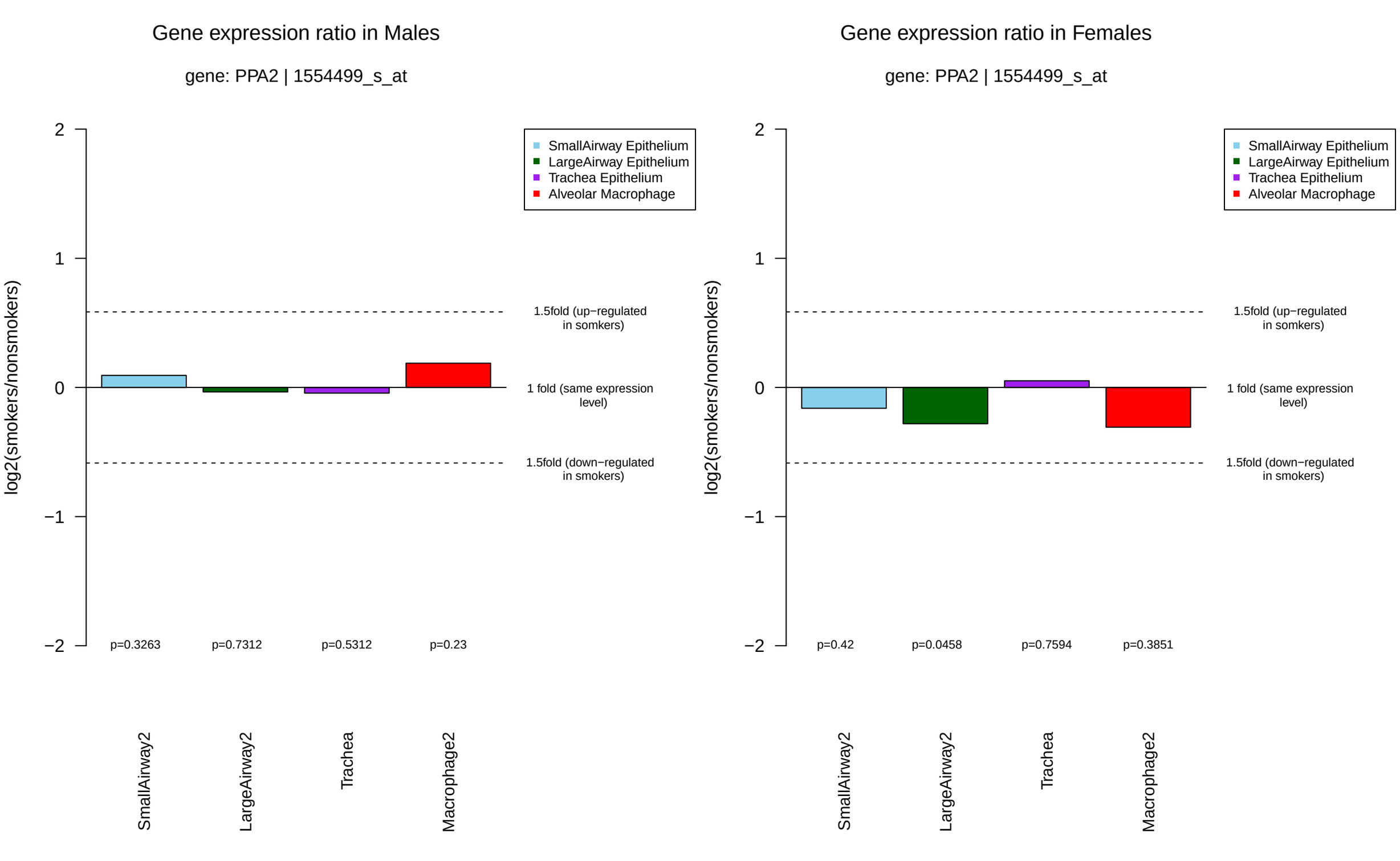
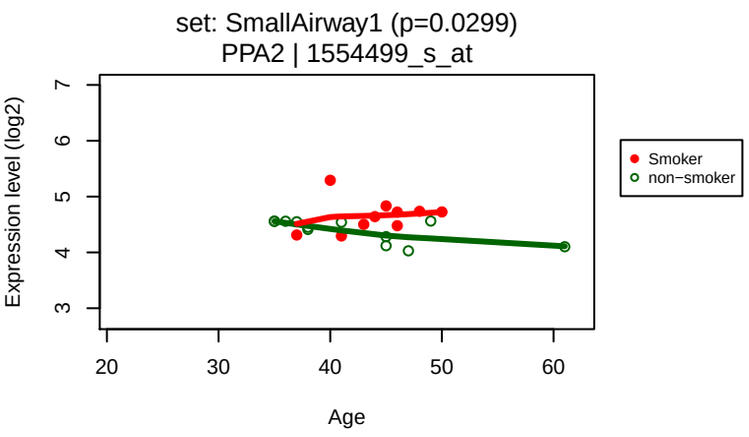
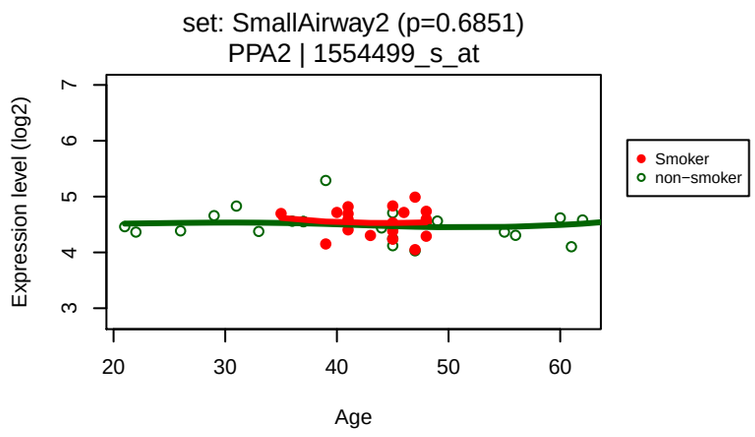
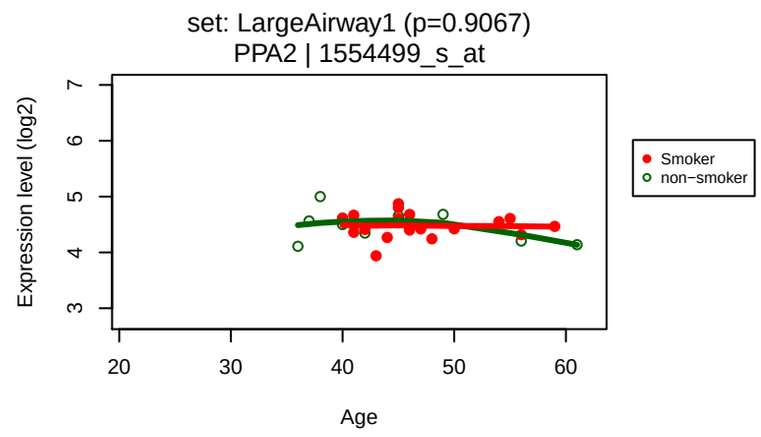

Gene expression of smokers and nonsmokers
gene: PPA2 | 1554499_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0299
p=0.6851
p=0.485
p=0.9067
p=0.1249
p=0.6403
p=0.3399
p=0.9172
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

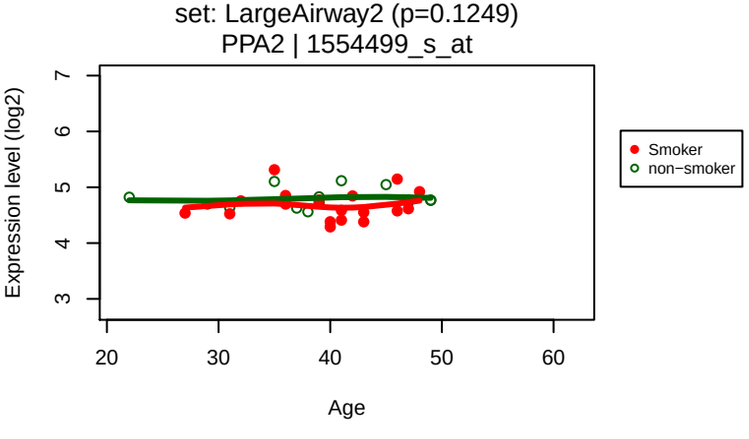

Gene expression ratio (smokers/nonsmokers)
gene: PPA2 | 1554499_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0299
p=0.6851
p=0.485
p=0.9067
p=0.1249
p=0.6403
p=0.3399
p=0.9172
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

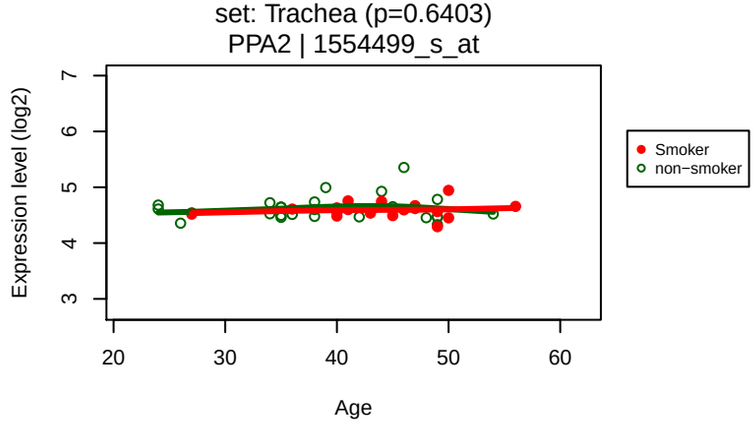

Gene expression ratio in Males
gene: PPA2 | 1554499_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3263
p=0.7312
p=0.5312
p=0.23
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPA2 | 1554499_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.42
p=0.0458
p=0.7594
p=0.3851
SmallAirway2
LargeAirway2
Trachea
Macrophage2

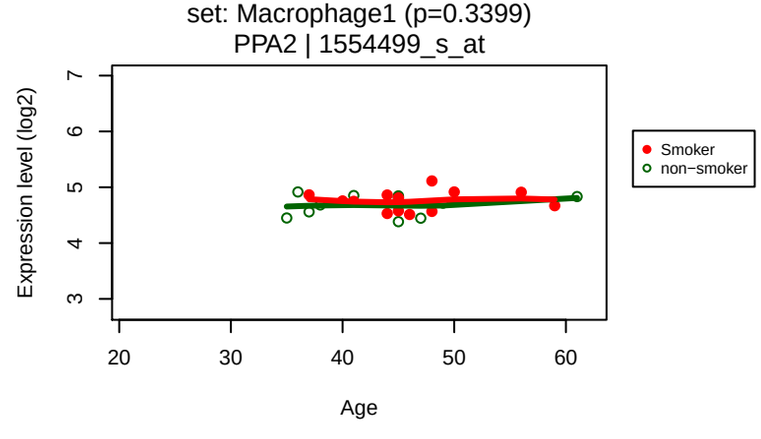

set: SmallAirway1 (p=0.0299)
PPA2 | 1554499_s_at
Expression level (log2)
Age
Smoker
non−smoker

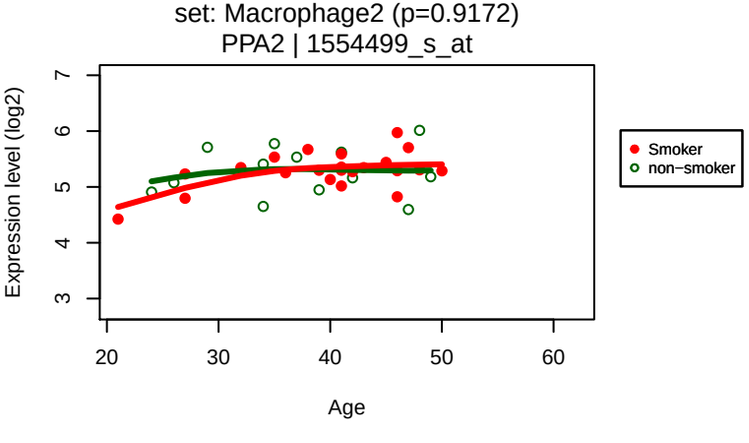

set: SmallAirway2 (p=0.6851)
PPA2 | 1554499_s_at
Expression level (log2)
Age
Smoker
non−smoker

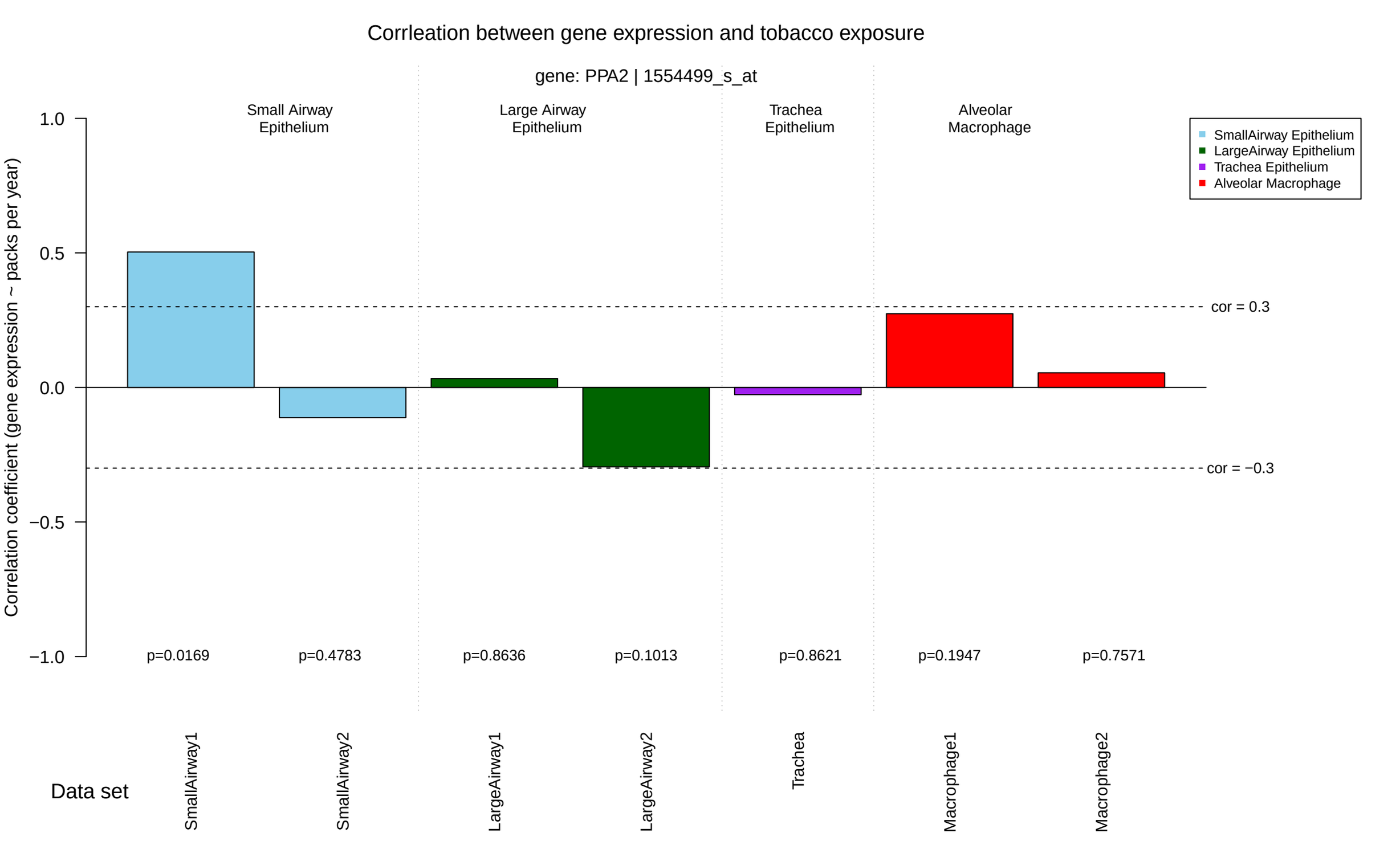

set: LargeAirway1 (p=0.9067)
PPA2 | 1554499_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1249)
PPA2 | 1554499_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6403)
PPA2 | 1554499_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3399)
PPA2 | 1554499_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9172)
PPA2 | 1554499_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPA2 | 1554499_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0169
p=0.4783
p=0.8636
p=0.1013
p=0.8621
p=0.1947
p=0.7571
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2